

Java Getter and Setter

What are getter and setter?

In Java, getter and setter are two conventional methods that are used for retrieving and updating value of a variable.

The following code is an example of simple class with a private variable and a couple of getter/setter methods:

```
public class A {  
    private int number;  
  
    public int getNumber() {  
        return number;  
    }  
  
    public void setNumber(int num) {  
        this.number = num;  
    }  
}
```

The class declares a private variable, number. Since number is private, code from outside this class cannot access the variable directly, like this:

```
A obj = new A();  
obj.number = 10;    // compile error, since number is private  
int num = obj.number; // same as above
```

Instead, the outside code have to invoke the getter, getNumber() and the setter, setNumber() in order to read or update the variable, for example:

```
A obj = new A();  
obj.setNumber(10); // OK  
int num = obj.getNumber(); // fine
```

So, a setter is a method that updates value of a variable. And a getter is a method that reads value of a variable.

Why getter and setter?

By using getter and setter, the programmer can control how his important variables are accessed and updated in a correct manner, such as changing value of a variable within a specified range.

setter and getter methods protect a variable's value from unexpected changes by outside world - the caller code.

When a variable is hidden by private modifier and can be accessed only through getter and setter, it is encapsulated. Encapsulation is one of the fundamental principles in object-oriented programming (OOP), thus implementing getter and setter is one of ways to enforce encapsulation in program's code.

Some frameworks such as Hibernate, Spring, Struts... can inspect information or inject their utility code through getter and setter. So providing getter and setter is necessary when integrating your code with such frameworks.

Naming convention for getter and setter

The naming scheme of setter and getter should follow Java bean naming convention as follows:

getXXX() and setXXX()

where XXX is name of the variable. For example with the following variable name:

```
private String name;
```

then the appropriate setter and getter will be:

```
public void setName(String name) { }
```

```
public String getName() { }
```

Example 01

public class Adhar

```
{  
    private String name;  
    private long mobno;  
    private int id;  
    private float hight;  
    private double salary;  
    private String addr;  
  
    public void setName(String name) {  
        this.name = name;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public void setMobno(long mobno) {  
        this.mobno = mobno;  
    }  
  
    public long getMobno() {  
        return mobno;  
    }  
  
    public void setId(int id) {  
        this.id = id;  
    }  
  
    public int getId() {  
        return id;  
    }  
  
    public void setHight(float hight) {  
        this.hight = hight;  
    }  
  
    public float getHight() {  
        return hight;  
    }  
  
    public void setSalary(double salary) {  
        this.salary = salary;  
    }  
}
```

```
public double getSalary() {  
    return salary;  
}  
  
public void setAddr(String Addr) {  
    this.addr = Addr;  
}  
  
public String getAddr() {  
    return addr;  
}  
}  
  
public class AdharTest  
{  
    public static void main(String args[]) {  
        Adhar a = new Adhar();  
        a.setName("ABC");  
        a.setMobno(95959595951);  
        a.setId(14);  
        a.setHight(5.15f);  
        a.setSalary(25000);  
        a.setAddr("Pune");  
  
        System.out.println(a.getName());  
        System.out.println(a.getMobno());  
        System.out.println(a.getId());  
        System.out.println(a.getHight());  
        System.out.println(a.getSalary());  
        System.out.println(a.getAddr());  
    }  
}
```